

CS6101 PA3 Report

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1 Stage 1: Retriever

1.1 Loss Function

The score function is:-

$$s(q, d) = \sum_{t \in q \cap d} \text{impact}(t, d)$$

The loss function is:-

$$\mathcal{L}(q, d^+, d^-) = -\log \frac{\exp(s(q, d^+))}{\exp(s(q, d^+)) + \exp(s(q, d^-))}$$

1.2 Dataset Creation and Preprocessing

We are given the mapping from queries to relevant documents in the BEIR datasets. So for each pair of query and relevant document, we randomly choose a document other than the relevant documents as the irrelevant document, irrespective of the dataset (quora or trec-covid).

Throughout the whole assignment, the BERT embeddings for queries and documents have been clipped to 160 tokens.

1.3 Train/Test Statistics

Dataset	Train	Test
quora	10000	5000
trec-covid	42	8

Table 1: Number of queries in each dataset

We have 61337 rows of training data, i.e., tuples of (query, good doc, bad doc). All models have been trained for 2 epochs collectively on the quora and trec-covid dataset.

1.4 Results

1.4.1 Baseline (BM25)

Dataset	Recall@100	Success@100
quora	0.9566	0.9744
trec-covid	0.0383	1.0000

1.4.2 Only MLP Training

Dataset	Recall@100	Success@100
quora	0.7661	0.8068
trec-covid	0.0275	1.0000

1.4.3 MLP+BERT Training (Extra Credit 1)

Dataset	Recall@100	Success@100
quora	0.9104	0.9384
trec-covid	0.0251	1.0000

1.4.4 MLP+BERT Training with Doc Expansion (Extra Credit 3)

Dataset	Recall@100	Success@100
quora	0.8865	0.9168
trec-covid	0.0223	1.0000

1.4.5 Quantized Retrieval (Extra Credit 2)

Model	Dataset	Recall@100	Success@100
Only MLP	quora	0.7961	0.8384
	trec-covid	0.0236	1.0000
MLP+BERT	quora	0.7807	0.8184
	trec-covid	0.0278	1.0000
MLP+BERT with Doc Expansion	quora	0.7311	0.7700
	trec-covid	0.0257	1.0000

1.5 Extra Credit Implementation Details

1.5.1 Extra Credit 1

We make the whole BERT model trainable. We notice a significant increase in Recall@100 and Success@100 for the quora dataset.

- Number of parameters in MLP - 197121
- Number of parameters in the whole model (BERT+MLP) - 109711419

1.5.2 Extra Credit 2

We do a linear quantization scheme to the range $[1, 2^8 - 1]$, i.e., we store our scores in an 8-bit scheme.

1.5.3 Extra Credit 3

We use the `castorini/doc2query-t5-base-msmarco` model for document expansion.

Original Document	Expansion Terms
Which one dissolve in water quickly sugar, salt, methane and carbon di oxide?'	which one dissolves in water quickly sugar
What career advice would you give to someone who wants to be a financial analyst?	how to become a financial analyst?

2 Stage 2: Reranker

For this stage, we are using the MLP+BERT with Doc Expansion model. We start with initial values:-

- $\lambda_T = 0.8$
- $\lambda_O = 0.1$
- $\lambda_U = 0.1$

2.1 Function Definitions

- $f_T(q_i, D) = \frac{1 + \cos(q_i, D)}{2}$
- $f_O(q_i, q_{i+1}, D) = \frac{1 + \cos(\frac{q_i + q_{i+1}}{2}, D)}{2}$
- $f_U(q_i, q_j, D) = \frac{1 + \cos(\frac{q_i + q_j}{2}, D)}{2}$, where $1 < |i - j| \leq \text{window_size}$

2.2 Results

Model	λ_T	λ_O	λ_U	Dataset	MAP	NDCG@10
Baseline (Stage 1)	-	-	-	quora	0.5471	0.5884
				trec-covid	0.0114	0.5368
Window Size=5	0.55	0.1	0.35	quora	0.6381	0.6830
				trec-covid	0.0135	0.6815
Window Size=10	0.85	0.1	0.05	quora	0.6340	0.6797
				trec-covid	0.0134	0.6758
Window Size=20	0.85	0.1	0.05	quora	0.6340	0.6797
				trec-covid	0.0133	0.6703

With the help of reranking, we observe a noticeable improvement compared to the Stage 1 retriever.

With increasing window size, the performance slightly decreases.